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NLP & LLM Terms Cheat sheet

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Natural Language Processing.(NLP).

NLP is a field of AI focused on enabling computers to understand, interpret, and generate human language. It encompasses a wide range of tasks, such as text classification, sentiment analysis, and machine translation.

Large Language Models (LLM).

LLMs, a subset of NLP, are advanced AI models designed to understand and generate human language using deep learning techniques, particularly transformers. Examples include GPT-4 and BERT, which perform tasks like text generation and translation with high proficiency.

Now let's see the common terms in it





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Terms in NLP (10/23)

- **Tokenization:** Splitting text into individual words or subwords.
- **Lemmatization:** Reducing words to their base or root form.
- **Stemming:** Trimming words to their base form by removing suffixes.
- **Part-of-Speech Tagging (POS):** Assigning parts of speech to each word in a sentence.
- **Named Entity Recognition (NER):** Identifying and classifying entities in text (e.g., names, dates).
- **Sentiment Analysis:** Determining the emotional tone of text.
- **Stop Words:** Common words (e.g., "and", "the") that are often removed from text during preprocessing.
- **Word Embeddings:** Representing words as vectors in a continuous vector space.
- **Bag of Words (BoW):** Representing text as the frequency of each word within it.
- **TF-IDF (Term Frequency-Inverse Document Frequency):** Reflecting the importance of a word in a document relative to a corpus.



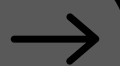


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Terms in NLP (23/23)

- **n-grams:** Sequences of 'n' consecutive words in a text.
- **Dependency Parsing:** Analyzing grammatical structure and relationships between words.
- **Coreference Resolution:** Identifying when different expressions refer to the same entity.
- **Word Sense Disambiguation:** Determining which meaning of a word is used in a context.
- **Language Modeling:** Predicting the next word in a sequence of words.
- **Semantic Role Labeling:** Assigning roles to words in a sentence according to their relationships.
- **Text Summarization:** Creating a concise summary of a larger text.
- **Text Classification:** Categorizing text into predefined groups.
- **Text Generation:** Automatically creating text based on a given input.
- **Topic Modeling:** Identifying topics present in a collection of documents.
- **Parsing:** Analyzing the syntactic structure of sentences.
- **Sentence Segmentation:** Dividing text into individual sentences.
- **Language Detection:** Identifying the language of a given text.





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Terms in LLM (10/25)

- **Transformer:** A neural network architecture designed for handling sequences, central to many LLMs.
- **BERT (Bidirectional Encoder Representations from Transformers):** A pre-trained transformer model for NLP tasks that considers context from both directions.
- **GPT (Generative Pre-trained Transformer):** A transformer-based model focused on text generation.
- **Fine-tuning:** Adapting a pre-trained model to a specific task or dataset.
- **Pre-training:** Training a model on a large dataset to learn general language patterns.
- **Masked Language Model:** A training objective where certain words in a sentence are masked, and the model predicts them.
- **Next Sentence Prediction:** A task where the model predicts if one sentence logically follows another.
- **Attention Mechanism:** A technique that allows models to focus on relevant parts of the input sequence.
- **Self-Attention:** A mechanism within transformers where each word in a sentence pays attention to other words.
- **Context Window:** The range of text a model can consider when making predictions.





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Terms in LLM (20/25)

- **Embedding Layer:** The layer of a neural network that converts text to vectors.
- **Layer Normalization:** A technique to stabilize and speed up the training of deep neural networks.
- **Beam Search:** A search algorithm used to generate sequences by exploring multiple possible outputs.
- **Recurrent Neural Network (RNN):** A type of neural network designed for sequential data, though less used in modern LLMs.
- **Long Short-Term Memory (LSTM):** A type of RNN that can capture long-term dependencies.
- **Bidirectional RNN:** An RNN variant that processes data in both forward and backward directions.
- **Sequence-to-Sequence (Seq2Seq):** A model framework for tasks where input and output are sequences, like translation.
- **Prompt:** Input text given to a language model to elicit a response.
- **Zero-Shot Learning:** Performing a task without any specific training examples for that task.
- **Few-Shot Learning:** Learning to perform a task given only a few examples.





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Terms in LLM (25/25)

- **Parameter:** The weights and biases in a neural network that are learned during training.
- **Overfitting:** When a model learns noise in the training data instead of the actual patterns, performing poorly on new data.
- **Underfitting:** When a model is too simple to capture the underlying patterns in the data.
- **Transfer Learning:** Applying knowledge from one task to improve performance on another related task.
- **Inference:** The process of making predictions using a trained model.
- **Encoder:** The encoder processes the input text into a fixed representation,
- **Decoder:** The decoder generates the output text from this representation.





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